Complex Numbers

Section 1: Cartesian Form & Complex Arithmetic

1 Given $z = 3 - 4i$, compute $z \cdot z^*$ where z^* is the complex conjugate of z.

a. 25

b. 7

c. -7

d. $25 - 24i$

2 Express $\dfrac{5 + 2i}{2 - i}$ in the form $a + bi$.

a. $\dfrac{8}{5} + \dfrac{9}{5}i$

b. $\dfrac{12}{5} + \dfrac{9}{5}i$

c. $8 + 9i$

d. $\dfrac{8}{3} + \dfrac{9}{3}i$

3 Find the two square roots of $-5 + 12i$.

a. $\pm(2 + 3i)$

b. $\pm(3 + 2i)$

c. $\pm(2 - 3i)$

d. $\pm(3 - 2i)$

Section 2: Modulus-Argument Form

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4 Find the modulus and principal argument of $z = -1 + i\sqrt{3}$.

a. $r = 2, \theta = \frac{2\pi}{3}$

b. $r = 2, \theta = \frac{\pi}{3}$

c. $r = 4, \theta = \frac{2\pi}{3}$

d. $r = 2, \theta = -\frac{2\pi}{3}$

5 Given $z_1 = 3(\cos(\pi/4) + i\sin(\pi/4))$ and $z_2 = 2(\cos(\pi/6) + i\sin(\pi/6))$, find $\arg(z_1 z_2)$.

a. $\frac{5\pi}{12}$

b. $\frac{\pi}{12}$

c. $\frac{\pi}{24}$

d. $\frac{7\pi}{12}$

6 Express $z = 4e^{i\pi/3}$ in Cartesian form $x + iy$.

a. $2 + 2i\sqrt{3}$

b. $2\sqrt{3} + 2i$

c. $2 - 2i\sqrt{3}$

d. $4 + 4i\sqrt{3}$

Section 3: Loci in the Argand Plane

7 Describe the locus of points represented by $|z - 3 - 4i| = 5$ in the complex plane.

a. A circle with center $(3, 4)$ and radius 5

b. A circle with center $(-3, -4)$ and radius 5

c. A circle with center $(3, 4)$ and radius 25

d. A straight line perpendicular to $3 + 4i$

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8 Describe the locus of points satisfying $|z-2| = |z-6i|$.

a. Perpendicular bisector of the segment joining $(2, 0)$ and $(0, 6)$

b. Circle passing through $(2, 0)$ and $(0, 6)$

c. Ellipse with foci at $(2, 0)$ and $(0, 6)$

d. Line segment joining $(2, 0)$ and $(0, 6)$

9 Describe the geometric locus defined by $\arg(z-2) = \dfrac{\pi}{4}$.

a. A half-line starting from $(2, 0)$ making an angle of $\pi/4$ with the positive real axis

b. A full line through $(2, 0)$ with gradient 1

c. A circle of radius 2

d. A ray towards the origin

Section 4: Roots of Polynomials & Conjugate Pairs

10 Given that $z = 1 + 2i$ is a root of $z^3 - 3z^2 + 7z - 5 = 0$, what are the other two roots?

a. $z = 1 - 2i$ and $z = 1$

b. $z = -1 - 2i$ and $z = 1$

c. $z = 1 - 2i$ and $z = 5$

d. $z = -1 + 2i$ and $z = -1$

11 Find a quadratic equation with real coefficients having root $z = 4 - 3i$.

a. $z^2 - 8z + 25 = 0$

b. $z^2 + 8z + 25 = 0$

c. $z^2 - 8z + 7 = 0$

d. $z^2 - 6z + 25 = 0$

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12 Find the least value of $|z|$ for points z satisfying $|z - 5 - 12i| \leq 3$.

a. 10

b. 13

c. 16

d. 7

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